Alberta Public Health Disease Management Guidelines

West Nile Virus

Includes:
West Nile Neurological Syndrome
West Nile Non-Neurological Syndrome and
West Nile Asymptomatic Infection

This publication is available online at https://open.alberta.ca/publications/west-nile-virus

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Primary and Preventative Health Services

West Nile Virus | Primary and Preventative Health Services, Government of Alberta

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Revisions

Revision Date	Document Section	Description of Revision
July 2025	Guideline	• Entire guideline updated to align with current information. • Epidemiology section title changed to “Clinical Assessment and Epidemiology.”
	Case Definition	• Updated case definitions based on revised 2024 national case definitions • Removed suspect case definitions as they were only provided to assist with case classification and were not notifiable • West Nile Asymptomatic Infection (WNAI) – updated information on blood operator testing results (confirmed and probable)

Case Definition

West Nile Neurological Syndrome (WNNS)

Confirmed Case

Clinical criteria[(A)] and laboratory confirmation of infection:

- A significant (i.e., fourfold or greater) change in West Nile virus (WNV) neutralizing antibody titres (i.e., plaque reduction neutralization tests [PRNT])[(B)] in paired acute and convalescent sera, or cerebrospinal fluid (CSF) sample;

OR

- Isolation of WNV from, or demonstration of WNV antigen in tissue, or detection of WNV-specific genomic sequences (e.g., polymerase chain reaction [PCR]) in tissue, blood, CSF or other body fluids;

OR

- Demonstration of WNV antibodies in a single serum or CSF sample using a WNV IgM enzyme-linked immunoassay (ELISA or EIA), confirmed by the detection of WNV neutralizing antibodies using a PRNT[(B)] (CSF or acute or convalescent serum).

Probable Case

Clinical criteria[(A)] and ONE or more of the following serology test results:

- WNV IgM positive in a single serum[(C)] or CSF specimen without confirmatory neutralization serology (i.e., PRNT);

OR

- WNV IgM positive and WNV IgG positive (low avidity), in a single serum[(C)] or CSF specimen;

OR

- A significant (i.e., fourfold or greater) change in WNV IgG, in paired acute and convalescent sera, or by seroconversion[(D)], using enzyme immunoassay (EIA) without confirmatory neutralization serology (i.e., PRNT);

[(A)] **Clinical criteria**:

New onset of **AT LEAST ONE** of the following: encephalitis, viral meningitis, acute flaccid paralysis*, or other neurological syndromes (e.g., cranial nerve palsies),

AND

- History of exposure when and where WNV transmission is present historically, or could be present, or history of travel to an area with confirmed WNV activity in birds, horses, other mammals, mosquitoes, or humans.

OR

- History of exposure to an alternate mode of transmission (laboratory-acquired, in utero, receipt of blood components, organ/tissue transplant, and possibly via breast milk).

NOTE: Fever may not be present in all cases therefore not a requirement for each case.
* For the purpose of WNNS classification, muscle weakness is characterized by severe, non-transient and prolonged symptoms.

Refer to the National Case Definition for West Nile Virus for further details and considerations.

[(B)] **PRNT:** Confirm the first 3–5 locally acquired cases annually; subsequent probable cases may be classified as "confirmed" for surveillance. PRNT on a follow-up serum sample is recommended alongside the initial CSF sample.
Cross-reactivity: Travel and vaccination history should guide additional flavivirus testing (e.g., Dengue, St. Louis encephalitis). PRNT is needed to confirm WNV if there's concern about cross-reactivity from other flaviviruses or recent flavivirus immunization.

[(C)] WNV IgM antibodies can persist for over a year, complicating the distinction between recent and past infections in endemic areas. A single serum IgM result should be confirmed with PCR, avidity or PRNT testing if possible.

[(D)] Seroconversion confirms current WNV infection and typically occurs 4–10 days post-infection. Collecting acute and convalescent sera helps avoid misinterpretation and identifies initial cases. Static antibody titres in paired sera may still indicate recent infection if seroconversion is missed.

West Nile Non-Neurological Syndrome (WN Non-NS)

Confirmed Case

Clinical criteria[E] and laboratory confirmation of infection:

- A significant (i.e., fourfold or greater) change in WNV neutralizing antibody titres (i.e., PRNT)[F] in paired acute and convalescent sera;

OR

- Isolation of WNV from, or demonstration of WNV antigen in tissue, or detection of WNV-specific genomic sequences (e.g., PCR) in tissue, blood, or other body fluids;

OR

- Demonstration of WNV antibodies in a single serum using a WNV IgM ELISA/EIA, confirmed by the detection of WNV neutralizing antibodies using a PRNT (acute or convalescent serum).

Probable Case

Clinical criteria[C] and ONE of the following serology results:

- WNV IgM positive in a single serum[G] without confirmatory neutralization serology (i.e., PRNT);

OR

- WNV IgM positive and WNV IgG positive (low avidity) in a single serum[G];

OR

- A significant (i.e., fourfold or greater) change in WNV IgG, in paired acute and convalescent sera, or by seroconversion[H], using enzyme immunoassay (EIA) without confirmatory neutralization serology (i.e., PRNT).

[E] **Clinical criteria**:

- **New onset of AT LEAST TWO** of the following:
  - Arthralgia
  - Fatigue
  - Headache
  - Myalgia*
  - Rash (maculopapular or morbilliform)
  - Gastrointestinal manifestations (e.g., vomiting, nausea, diarrhea, abdominal pain)
  - Other (e.g., eye pain, chills, back pain)

AND

- History of exposure when and where WNV transmission is present historically, or could be present, or history of travel to an area with confirmed WNV activity in birds, horses, other mammals, mosquitoes, or humans.

  OR

- History of exposure to an alternate mode of transmission (e.g., laboratory-acquired, in utero, receipt of blood components, organ/tissue transplant, and possibly via breast milk)

* Myalgia or muscle weakness is characterized by mild, transient symptoms that are not associated with motor neuropathy.

NOTE: Fever may not be present in all cases and is therefore not a requirement for each case.

Refer to the National Case Definition for West Nile Virus for further details and considerations.

[F] **PRNT:** Confirm the first 3–5 locally acquired cases annually; subsequent probable cases may be classified as "confirmed" for surveillance.

Cross-reactivity: Travel and vaccination history should guide additional flavivirus testing (e.g., Dengue, St. Louis encephalitis). PRNT is needed to confirm WNV if there is concern about cross-reactivity from other flaviviruses or recent flavivirus immunization.

[G] WNV IgM antibodies can persist for over a year, complicating the distinction between recent and past infections in endemic areas. A single serum IgM result should be confirmed with PCR, avidity or PRNT testing if possible.

[H] Seroconversion confirms current WNV infection and typically occurs 4–10 days post-infection. Collecting acute and convalescent sera helps avoid misinterpretation and identifies initial cases. Static antibody titres in paired sera may still indicate recent infection if seroconversion is missed.

West Nile Asymptomatic Infection (WNAI)

NOTE: Asymptomatic testing should be limited to blood and organ donors as part of blood and organ donor screening programs.

Confirmed Case

Absence of clinical criteria and laboratory confirmation of infection:

- A significant (i.e., fourfold or greater) change in WNV neutralizing antibody titres (i.e., PRNT) in paired acute and convalescent sera;

OR

- Isolation of WNV from, or demonstration of WNV antigen in tissue, or detection of WNV-specific genomic sequences (e.g., PCR) in tissue, blood, CSF or other relevant body fluids;

OR

- Demonstration of WNV antibodies in a single serum using a WNV IgM ELISA/EIA, confirmed by the detection of WNV neutralizing antibodies using a PRNT (acute or convalescent serum);

OR

- One of the following from a blood operator (e.g., Canadian Blood Services or Hema-Quebec) in Canada:[C]
  - Demonstration of Japanese encephalitis (JE) serocomplex-specific genomic sequences in blood by nucleic acid amplification test (NAAT) confirmed by WNV-specific NAAT or sequencing,

  OR
  - Demonstration of WNV-specific genomic sequences in blood by NAAT.

Probable Case

Absence of clinical criteria and ONE of the following laboratory serology results:

- A significant (i.e., fourfold or greater) change in WNV IgG, in paired acute and convalescent sera, or by seroconversion, using enzyme immunoassay (EIA) without confirmatory neutralization serology (i.e., PRNT);

OR

- Demonstration of JE serocomplex-specific genomic sequences in blood by NAAT by blood operators (e.g., Canadian Blood Services or Hema-Quebec) in Canada when WNV-specific NAAT was not performed.

[C] **Asymptomatic blood donors** are screened for WNV by blood operators using a nucleic acid amplification test (NAAT). Positive results are reported to Alberta Health. Blood operators may use a WNV-specific NAAT or a NAAT that detects viruses in the Japanese encephalitis serocomplex, including WNV. Subsequent confirmatory tests (WNV-specific NAAT) may be performed by ProvLab or National Microbiology Laboratory (NML).

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Zone Medical Officer of Health (MOH) (or designate), of all confirmed and probable WNNS, WN Non-NS and WNAI cases by mail, fax or electronic submission within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic submission within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- Zone MOH (or designate).

Primary Care Alberta and Indigenous Services Canada (ISC)

- The Zone MOH (or designate) where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable WNNS, WN Non-NS and WNAI cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country cases, the Zone MOH (or designate) should forward the following information to the CMOH (or designate) by phone, fax or electronic submission within 48 hours (two business days):
  - name,
  - date of birth,
  - out-of-province health care number,
  - out-of-province/country address and phone number,
  - positive laboratory report, and
  - other relevant clinical/epidemiological information.

Clinical Assessment and Epidemiology

Etiology

West Nile Virus (WNV) belongs to a family of viruses called *Flaviviridae.*[1,2] It is a member of the Japanese encephalitis (JE) virus complex that includes St. Louis encephalitis.[1]

Clinical Presentation

Most WNV infections (70–80%) are predominantly asymptomatic.[2–5] Approximately 20% of individuals may experience West Nile Non-Neurological Syndrome (WN Non-NS), which commonly manifests as fever, headache, myalgia, arthralgia, gastrointestinal tract symptoms and transient rash.[2–5] While most individuals with WN Non-NS recover completely, some may suffer from lingering symptoms such as fatigue, malaise and weakness for several months.[1,2,6]

Less than 1% of WNV-infected individuals develop severe neurological disease (i.e., WNNS), which can include meningitis, encephalitis, or acute flaccid paralysis.[2–4,6] Those recovering from WNNS may experience enduring neurological deficits.[1,2,6]

The case fatality rate for WNNS is approximated at 10%, with higher mortality rates observed in cases of WNV encephalitis or acute flaccid paralysis compared to WNV meningitis.[1–3,6]

Diagnosis

The diagnosis of WNV is often based on a high index of suspicion and obtaining the results of specific laboratory tests. The presence of WNV enzootic activity or human cases increases the likelihood of infection in the population.

Immunocompromised individuals may have delayed or absent antibody response, making serologic diagnosis challenging.[7] They may experience prolonged viremia, so viral detection methods may be helpful in diagnosis.[7] WNV diagnostic test criteria for these individuals should be discussed with the Microbiologist-on-Call (MOC).Travel and immunization history should guide additional flavivirus testing (e.g., Dengue, St. Louis encephalitis).[7] PRNT is needed to confirm WNV if there is concern about cross-reactivity from other flaviviruses or recent flavivirus immunization.[7]

See the Alberta Precision Laboratories (ProvLab) website for up-to-date information on WNV laboratory testing, and review the interpretation of WNV lab results table to support diagnosis.

Treatment

- There is no specific treatment, medication or cure for WNV.[1–3,8]
- Treatment for WNV is supportive, often requiring hospitalization for those with severe disease and may involve intravenous fluids, respiratory support, and the prevention and management of secondary infections.[2,3,8]

Reservoir

Birds are the primary reservoir for WNV.[2,3,9]

Transmission

The most common mode of transmission is through the bite of an infected mosquito (dawn and dusk feeder).[2,3,9] Transmission can occur through blood transfusion or transplanted organs from an infected donor.[2,3,9] There have also been rare reports of in-utero infection, transmission through breast milk, and infections in laboratory workers who handle infected specimens.[2] While viremia may be sufficient for rare human-to-human transmission through routes such as blood transfusion or organ transplant, it is typically not high enough to infect mosquitoes or for further transmission to humans.[2,3,9]

Incubation Period

Symptoms usually develop within 2 to 6 days but can range from 2 to 14 days and up to several weeks in immunocompromised individuals.[1–3,10]

Period of Communicability

Humans can be communicable (see Transmission) during the viremic phase which begins a few days before to a few days after the onset of clinical illness.[11]

Host Susceptibility

- Susceptibility is universal.
- Risk of infection is dependent on exposure to infected vectors, environmental conditions, season and individual activities.[2]
- If infected, higher risk of severe illness is associated with:[2,3,6]
  - older age (> 50 years),
  - chronic renal disease,
  - Immunocompromised conditions,
  - history of alcohol abuse,
  - diabetes,
  - hypertension, and
  - organ transplant recipients

Incidence

WNV has been notifiable in Canada and Alberta since 2003. The risk of WNV infection is highest between mid-July and early September.[12] Surveillance of humans, birds, mosquitos, and/or horses may be conducted at federal and provincial levels to determine the risk and incidence of WNV. For national surveillance reports, refer to the Government of Canada.

In Alberta, human cases of WNV have predominantly been reported in the southeast area of the province; however, they may occur anywhere especially during a year with higher incidence. In addition, Albertans have also acquired WNV via travel to endemic locations outside of Alberta and Canada. For information on cases in Alberta, refer to the Interactive Health Data Application (IHDA)

Canadian Blood Services (CBS)

CBS routinely screens for WNV in all donated specimens. Information on WNV incidence and surveillance in Canada's blood supply can be reviewed on the CBS website.

Organ and Tissue Donors

In Alberta, organ/tissue donors are tested for WNV by the ProvLab. Between 500 and 600 organs/tissue specimens are tested each year in Alberta, and none have tested positive since testing was first initiated in 2003.[13]

Public Health Management

Key Investigation

- Ensure the person meets case definition.
- Obtain a history of illness including the date of onset and signs and symptoms.
- Determine where the case may have been exposed in the two weeks prior to onset, taking into consideration the incubation period, reservoir, and mode of transmission, including a history of: [1–3,14]
  - Recent travel to, or residence in, an area with suitable seasonal and ecological conditions for WNV transmission (See Incidence),
  - Blood/blood component recipient (within eight weeks prior to symptoms onset),
  - Organ/tissue transplant recipient (within eight weeks prior to symptom onset),
  - Handling of sick or dead birds or animals,
  - Occupational unprotected exposure to the blood or body fluids containing WNV (e.g., laboratory worker, outdoor worker, bird/animal handler, health care worker),
  - Immunization for another flavivirus (i.e., yellow fever, Japanese encephalitis),
  - Possible transmission in utero or through breast milk in infant cases.

Management of a Case

- All cases should be advised:
  - Of disease transmission and prevention strategies.
  - To defer blood or organ donation for 56 days and consult Canadian Blood Services or the Transplant Program for additional guidance.[15]
- Infants born to persons infected with WNV during pregnancy should undergo clinical evaluation for WNV infection. Consultation with an infectious disease specialist is recommended.
  - The health benefits of breastfeeding are well established, and the risk for WNV transmission through breastfeeding is considered very low; therefore breastfeeding is *not* contraindicated.[16] Lactating persons experiencing illness or breastfeeding difficulties should consult their healthcare practitioner.

Management of Contacts

- Direct follow-up of contacts is not required. However, it may be prudent to identify others who may have been exposed to the same source as the case to find undiagnosed or unreported cases.
  - Educate identified contacts about WNV, including prevention methods, and advise them to seek medical attention if symptoms occur.

Preventive Measures

- Currently, there is no vaccine or medication to prevent WNV infection in humans.[1,2,17]
- The best way to prevent WNV is to avoid mosquito bites.[1–3,14,18]
- Inform the public about the following personal protective measures to reduce mosquito exposure:[1,3,14,18]
  - Avoid areas with high mosquito populations, especially at dawn and dusk, when possible.
  - Use mosquito repellent containing DEET, Picaridin or Permethrin, following Health Canada and Canadian Pediatric Society guidelines for age-appropriate use.
  - Wear permethrin-treated clothing when outdoors, along with long sleeves and pants, closed-toed footwear, and a hat.
  - Stay in well-screened or air-conditioned areas whenever possible.
  - Eliminate standing water around homes to reduce mosquito breeding sites (e.g., open-air containers such as flowerpots, bottles, used tires and rock pools).

- Ensure doors and windows are properly sealed to prevent mosquitoes from entering the home.
- Handling dead birds/animals:
  - Wear rubber gloves when handling dead birds or animals, wash gloved hands, then remove gloves and wash bare hands thoroughly,
  - Clean and cover any wounds, especially if there is any risk of contact with dead or sick birds or animals (e.g., hunting, veterinary care processing dead animals),
  - Report dead birds to the Canadian Wildlife Health Cooperative.

Additional Resources

Public Health Agency of Canada, West Nile virus website.

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